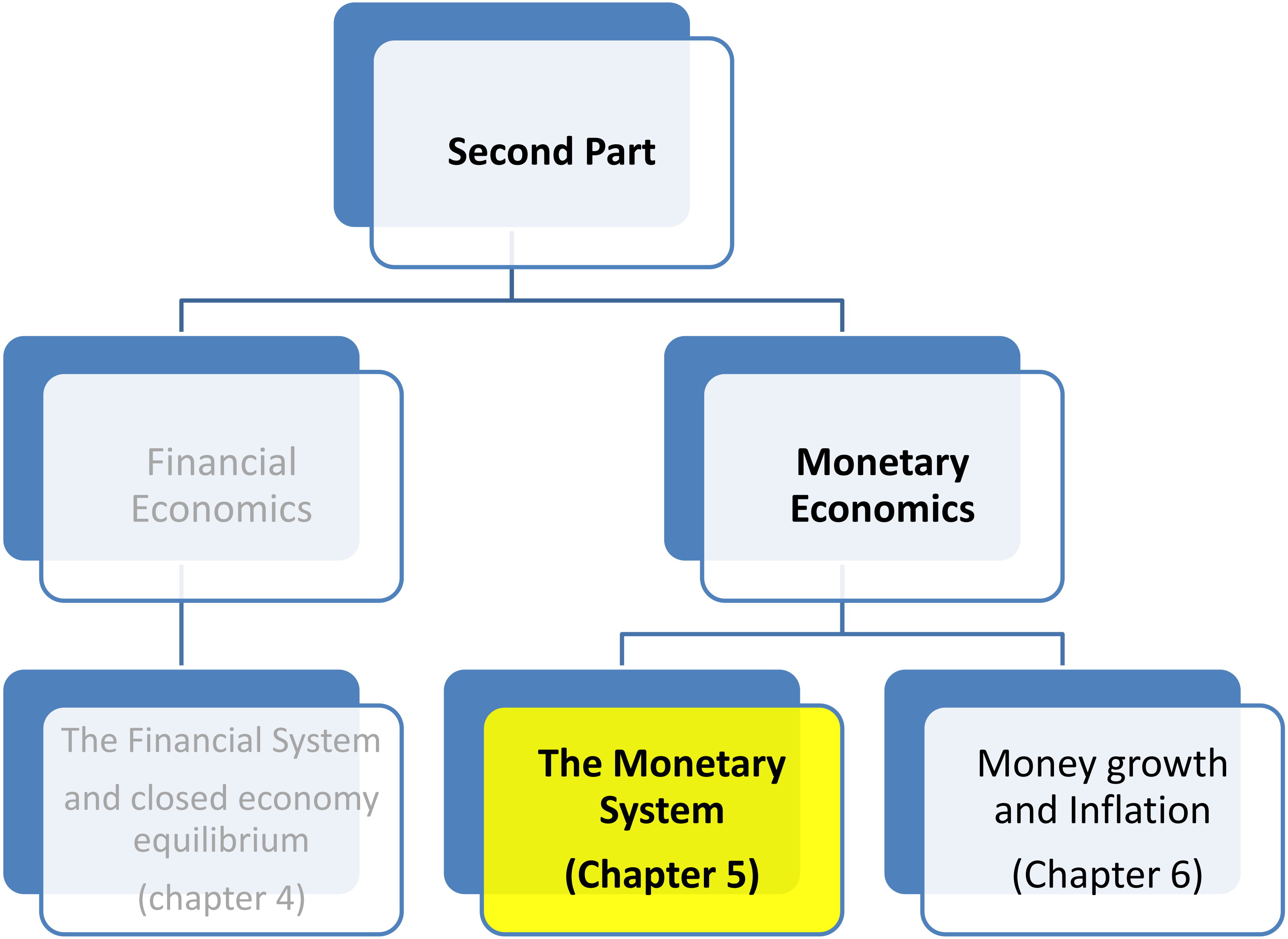
Introduction to Macroeconomics (S102016CR)

Spring Semester

https://moodle.unige.ch/course/view.php?id=6618

Marcelo Olarreaga, Office M5270


Second Part
Financial Economics
Monetary Economics
The Financial System and closed economy equilibrium (chapter 4)
The Monetary System (Chapter 5)
Money growth and Inflation (Chapter 6)

Week 5:
The Monetary System

What is money? Just a piece of paper? No, but where does the value of this piece of paper come from?

In this chapter, we focus on the functioning of the monetary system and the role played in terms of the money supply by:

a) Different types of money,
b) The Central Bank and its monetary policies
c) Commercial banks and their lending policies

It is important to understand the determinants of money supply because, as we will see later, it plays a crucial role in many macroeconomic variables, such as inflation, interest rates, production, and short-run unemployment.

In the next chapter, we will introduce money demand and the equilibrium in the market for money.

Chapter Contents

a) Definitions and the functions of money

b) The role of the Central Bank

c) The role of commercial banks

d) Summary

Chapter Contents

a) Definitions and the functions of money

b) The role of the Central Bank

c) The role of commercial banks

d) Summary

a. Definitions and the functions of money

- *Money* has three important functions:
  - Medium of exchange is an item that buyers give sellers when they want to purchase goods and services, i.e., anything readily acceptable as payment.
  - Unit of account is the yardstick people use to post prices and record debts
  - Store of value is an item people can use to transfer purchasing power from the present to the future.

- *Liquidity* is the ease with which an asset can be converted into the economy's medium of exchange. Money is the most liquid asset available. Other assets (such as stocks, bonds, and real estate) vary in their liquidity.

- When people decide in what forms to hold their wealth, they have to balance the *liquidity* of each possible asset against the asset's usefulness as a *store of value.*

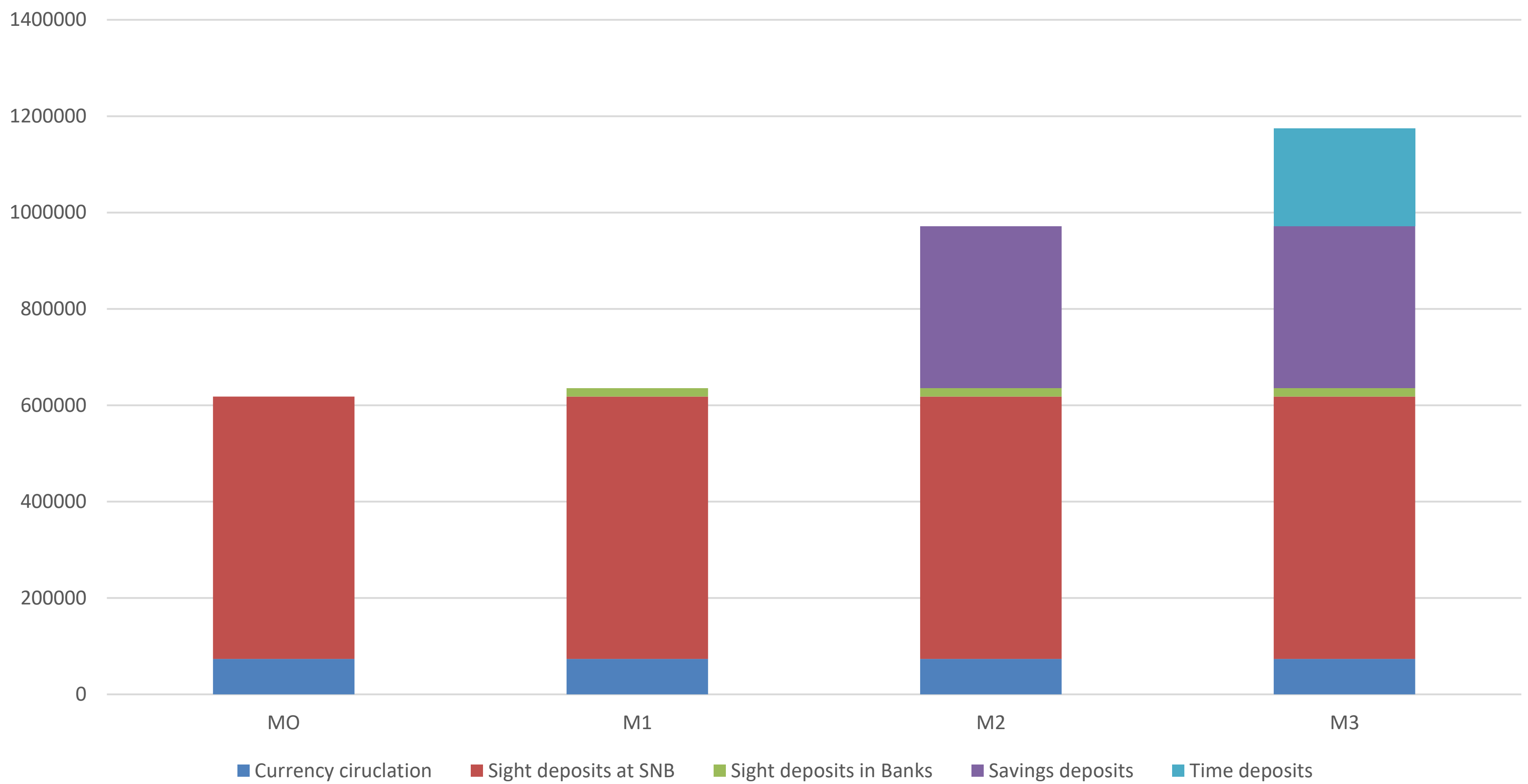
Types of money:

- *Commodity money* takes the form of a commodity with intrinsic value.
  - Examples: Gold, silver, cigarettes.
- *Fiat money* is used as money because of government decree.
  - It does not have intrinsic value.
  - Examples: Coins, currency, current account deposits.
  - Bitcoins are another example of fiat money without a decree

Money circulating in the economy

- *Currency* is the paper bills and coins in the hands of the public.
- *Demand/sight deposits* are balances in bank accounts that depositors can access on demand by writing a check or using a debit card.
- Currency plus sight deposits of commercial banks at the SNB is MO, or the monetary base
- The sum of currency and all sight deposits is called money supply M1
- Other definitions of money supply are less liquid and include savings accounts or time deposits (M2 and M3, respectively).

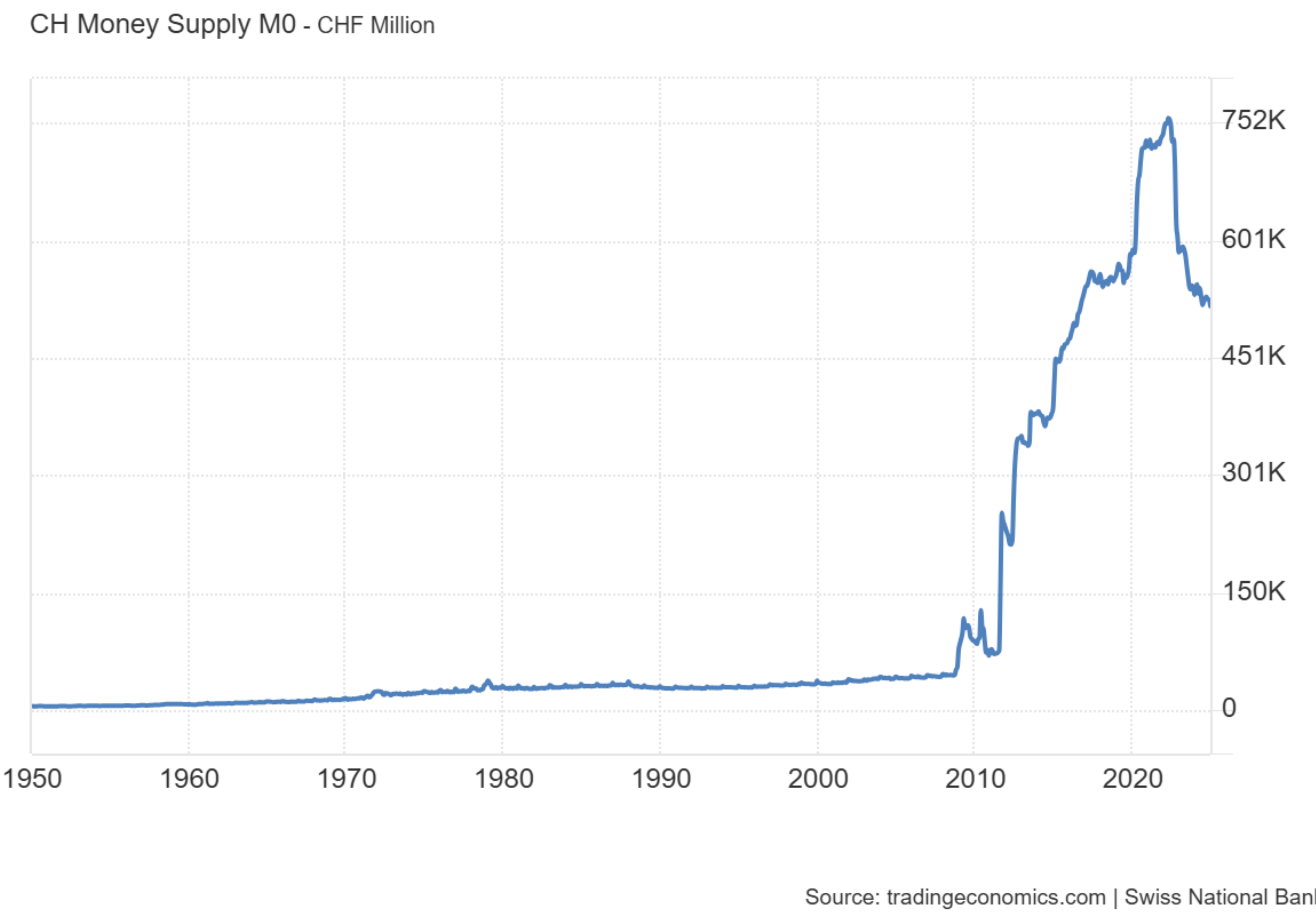
Definitions of money circulating in Switzerland (December 2024) – Millions of CHF



Source: SNB

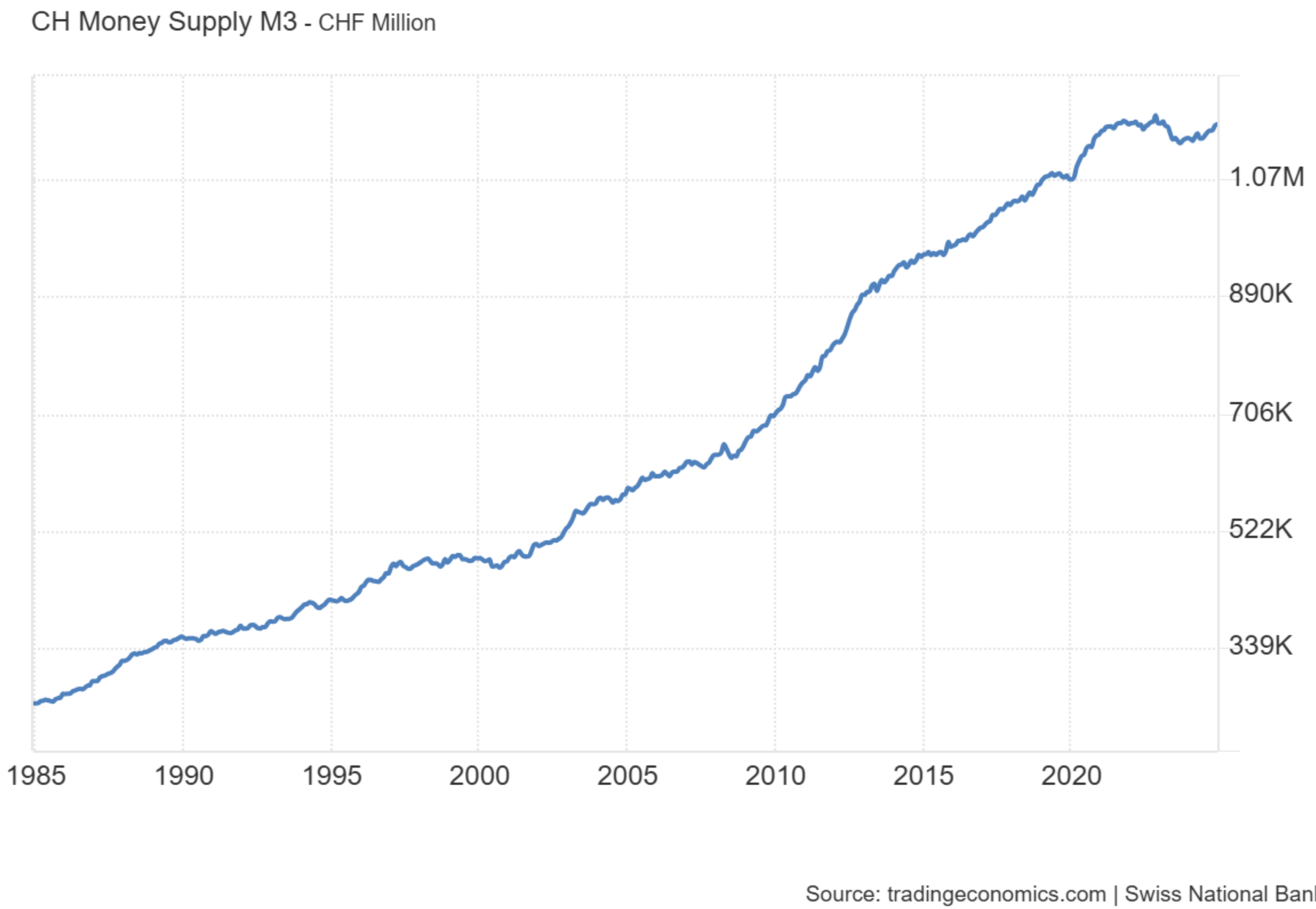
Liquidity declines as we go from M0 to M3. Savings can be accessed with three months notice, and term deposits depend on the maturity

Evolution of monetary base M0 in Switzerland



This puts in perspective the interventions of the SNB since the 2007-2009 financial crisis……

Evolution of money supply M2 in Switzerland



Steady increase of M3, and no significant break around the financial crisis or COVID due to substitution away from sight, savings and time deposits in commercial banks by sight deposits in SNB

Chapter Contents

a) Definitions and the functions of money

b) The role of the Central Bank

c) The role of commercial banks

d) Summary

b. The role of the Central Bank

- A *central bank* is an institution designed to oversee the banking system and regulate the quantity of money in the economy.

- Whenever an economy relies on fiat money, some agency must regulate the system.

- These are very important tasks because, as we will see later, money supply will determine inflation, interest rates, short-run unemployment, and production levels. It is essential to get it right….

- *Monetary policy* is the set of actions the central bank takes to affect the money supply.
  - Refinancing rate
  - Open market operations
  - Reserve requirements

The Refinancing or Repo

The *refinancing* is the interest rate the Central Bank lends on a short-term basis to the banking sector.

- Increasing the refinancing rate decreases the money supply as there is less demand for refinancing by commercial banks
- Decreasing the refinancing rate increases the money supply as commercial banks demand more refinancing.

– The refinancing rate is called the discount rate in the US

– Banks need to provide collateral if they want to benefit from Central Bank lending refinancing

– The collateral accepted is usually government bonds or bonds by large corporations on which there is little risk of default.

– At a specified later date, the transaction is reversed, which is why it is also called repurchased agreement or *repo*.

Open-Market Operations

- A central bank conducts *open-market operations* when it buys or sells bonds from or to the banking sector:
  - The money supply increases when the central bank buys bonds.
  - The money supply decreases when the central bank sells bonds.
- When the government purchases long-run or riskier financial assets (including toxic assets) from commercial banks and other private financial institutions to try to influence long-run yields, we call it "quantitative easing"
- The Fed, the ECB, and the Bank of England used quantitative easing during the financial crisis as short-term interest rates were near zero and traditional open-market operations were not achieving much.
- They usually work through a «reverse auction». The Central Bank announced a certain amount of assets to be purchased during a specific period, and banks and other private financial institutions bid an amount and a price for the assets it wanted to sell to the Central Bank.

Reserve Requirements

- *Reserve requirements* are regulations on the minimum reserves that banks must hold against deposits. Reserves are deposits that banks have received but have not loaned out.
- In a *fractional-reserve banking* system, banks hold a fraction of the money deposited as reserves and lend out the rest.
- They help avoid runs on banks.
- Central Banks can determine a minimum level of reserve for the system to be safe, and this will affect money supply:
  - Increasing the reserve requirement decreases the money supply.
  - Decreasing the reserve requirement increases the money supply.
- Central banks rarely change reserve requirements except for China, who has recently used them very often.
- Some, like the FED, the Bank of England or the Central Bank of Australia, have no minimum reserve requirements.
- The ECB has a reserve requirement of 1 percent.
- In Switzerland the SNB sets the minimum at 4 percent.

Different monetary policy objective(s)

ECB	US Federal Reserve	Swiss National Bank
Price stability is the primary objective as set in the Maastricht Treaty. The ECB has quantified this as medium-term inflation goal of "below but close to 2%"	Multiple objectives: to promote maximum employment, price stability, and moderate long-term interest rates. Price stability not defined, but widely viewed as 1-2% comfort zone (skewed toward upper portion) for core PCE inflation	Price stability is the main mandate interpreted as a range of 0% to 2% inflation per year. It is obliged by Constitution and statute to act in accordance with the interests of the country as a whole.

Source: Mongelli, Gerdesmeier & Roffia (2009)
http://www.voxeu.org/article/fed-eurosystem-and-bank-japan-similarities-and-differences

Chapter Contents

a) Definitions and the functions of money

b) The role of the Central Bank

c) The role of commercial banks

d) Summary

c. The role of commercial banks

- Banks can influence the quantity of demand deposits in the economy and the money supply by deciding how much money to lend beyond the minimum reserve requirements.

- When a bank makes a loan, the money supply (M1) increases.

- The money supply (M1) is affected by the amount deposited in banks and the amount banks loan.
  - Deposits into a bank are recorded as both *assets* and *liabilities*.
  - The fraction of total deposits a bank keeps as reserves is called the *reserve ratio*.
  - Loans become an asset to the bank.

- But as money is lent by one bank, it becomes a deposit in another bank, which will in turn loan part of this new deposit, continuing the process of money creation….This process is called the *money multiplier*

Money creation by commercial banks

- This T-Account shows a bank that…
  - accepts CHF100 deposit,
  - keeps a portion as reserves,
  - and lends out the rest.
  - It assumes a reserve ratio of 10%.
  - Notes that there is no wealth creation: assets= liabilities
- But this is only the beginning.
- The guy who took the loan will buy something, pay the seller and the seller will take this money to his bank (Second Swiss Bank).
- The money multiplier enters into action and commercial banks are creating money……

First National Bank

Assets		Liabilities	
Reserves	CHF 10	Deposits	CHF 100
Loans	CHF 90		
Total Assets	CHF 100	Total Liabilities	CHF 100

The Money Multiplier

First National Bank

Assets	Liabilities
Reserves CHF10	Deposits CHF100
Loans CHF90	
Total Assets CHF100	Total Liabilities CHF100

Second National Bank

Assets	Liabilities
Reserves CHF 9	Deposits CHF90
Loans CHF81	
Total Assets CHF90	Total Liabilities CHF90

Money Supply M1 = CHF 100 => CHF 190 => 271

The Money Multiplier

Second National Bank

Assets	Liabilities
Reserves CHF9	Deposits CHF90
Loans CHF81	
Total Assets CHF90	Total Liabilities CHF90

Third National Bank

Assets	Liabilities
Reserves CHF 8	Deposits CHF81
Loans CHF73	
Total Assets CHF81	Total Liabilities CHF81

Money Supply M1 = CHF 100 => CHF 190 => 271 => 344

Not theory, real world stuff…… The Third National Bank exists!!!!!

- And this keeps on going as more deposits will transform into more loans and, therefore, more deposits, which will increase the money supply

- At the end of the process, the total money supply is CHF 1000. How come? It can be shown that the money multiplier is the reciprocal of the reserve ratio:

$$m = 1/r$$

- With a reserve requirement, $r = 10\%$ or 0.1, the multiplier m is 10.

- And the money supply

$$MS = m * M0$$

- where M0 is the monetary base, i.e, banknotes and coins and deposits of commercial banks in the Central Bank.

- In this case M0=100, m=10, and MS=1000

Intuition for the money multiplier:

- Let’s think in reverse. If a bank has CHF1000 in deposits, and the reserve requirement is 10%, then it needs to have reserves for CHF 100, right?

- Thus, if the entire banking system has CHF 100 in banknotes that can be used as reserves, then if total reserves are 100, and if reserves requirements are 10%, this implies that the total amount of deposits in commercial banks will be CHF 1000.

- On Friday’s class we will show this more formally, but what you need to take away is twofold:
  - The money multiplier is m=1/r
  - Money supply is MS = m MO

- In explaining the money multiplier r, we have always assumed that households keep no liquidity and deposit all the currency they hold in commercial banks.

- But the reality is more subtle, and we do keep some currency outside the banking system (in our pocket or under the mattress).

- In that case, the money multiplier is smaller:

$$m = 1/(r+c)$$

- where c is the share of deposits held by households in cash (could also include banks if the cash they receive from the Central Bank is not lent out)
- Note that m declines with c
- 1/(r+c) can be seen as the multiplier effectively observed where 1/r is the maximum multiplier that we could observe in a world where households do not hold any cash….

Monetary policy: more an art than a science….

- It should be clear by now that a central bank's control of the money supply is not precise:

$$MS = m\ M0$$

- The Central Bank can only directly affect the monetary base (or M0), and the reserve requirements, but it can only very indirectly control what commercial banks and households are doing in the banking system.

- Essentially, a central bank must wrestle with two problems due to fractional-reserve banking.
  - The central bank does not control the amount of money households choose to hold as bank deposits.
  - The central bank does not control the amount of money that bankers choose to lend (beyond minimum reserve requirements).

Chapter Contents

a) Definitions and the functions of money

b) The role of the Central Bank

c) The role of commercial banks

d) Summary

d. Summary

- The term money refers to assets that people regularly use to buy goods and services.
- Money serves three economic functions: as a medium of exchange, a unit of account, and a store of value.
- Commodity money is money that has intrinsic value.
- Fiat money is money without intrinsic value.
- It is the function of a central bank to control the money supply through open-market operations, or by changing the refinancing rate, or by adjusting reserve requirements.
- When banks loan out their deposits, they increase the quantity of money in the economy.
- Because the central bank cannot control the amount bankers choose to lend or the amount households choose to deposit in banks, the central bank’s control of the money supply is imperfect.